higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PULP AND PAPERMAKING TECHNOLOGY
NQF LEVEL 3

14 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 False
1.2 True
1.3 True
1.4 False
1.5 False

(5 × 1) **[5]**

QUESTION 2

2.1 F
2.2 G
2.3 A
2.4 E
2.5 C

(5 × 1) **[5]**

QUESTION 3

3.1 Landfill
3.2 Process control
3.3 Shower
3.4 Channelling
3.5 Glossy paper

(5 × 1) **[5]**

QUESTION 4

4.1 4.1.1
- Receiving and sorting
- Slashing
- Debarking
- Chipping
- Chip screening
- Re-chipping
- Storing (7)

4.1.2 The process begins when the logs are brought into the woodyard for sorting and slashing.✓ The logs are debarked and sent to the chipper.✓ The bark from the debarking process is sent to refuse boilers or is thrown in a landfill depending on what the company chooses to do.✓ The chips from the chipper are sent for screening,✓ and any chips that are not the right size are sent to be re-chipped.✓ Finally, chips are stored so that they can be sent away to be pulped.✓ (6)

4.2 4.2.1 Bagasse washer
4.2.2 Depither
4.2.3 Pin drum feeder
4.2.4 Dewatering press

(4 × 1) (4)
[17]

QUESTION 5

5.1
- Thickness
- Tensile or stretch
- Roughness
- Brightness
- Tear
- Opacity
- Wax prick
- Density
- Spot or dirt count
- International graphite technique or printability test (Any 5 × 1) (5)

5.2 A – Storage bin
B – Primary pressure refiner
C – Cyclone
D – Latency chest
E – Thickener
F – Storage
G – Rejects system
H – Effluent (8)

5.3
- Water should be kept at a constant high temperature and at sufficient pressure.
- The surface of the stone must be cleaned properly.
- Water must flow constantly.
- The temperature of the grindstone must be controlled effectively.
- Water must not flood the grinding zone. (Any 4 × 1) (4)

[17]

QUESTION 6

6.1 Steaming makes the chips less brittle✓ and produces fewer shives and longer fibres.✓ (2)

6.2 Fibres dissolve with lignin during the Kraft process. (1)

6.3 Blow in a stream of steam at the top of the digester to blow the chips✓ and spread them more evenly in the digester.✓ Distribute the cooking liquor evenly through the chips.✓ (3)

6.4
- Variation in chip size
- Moisture content is low
- Liquid to wood ratio is low
- Low chemical concentration
- Short cooking time (5)

6.5
- Blow tank
- Continuous horizontal digester (2)

6.6
- Dark or burned pulp
- Low yield
- Large amount of fibre bundles and uncooked chips
- Large volume of red liquor produced (4)

[17]

QUESTION 7

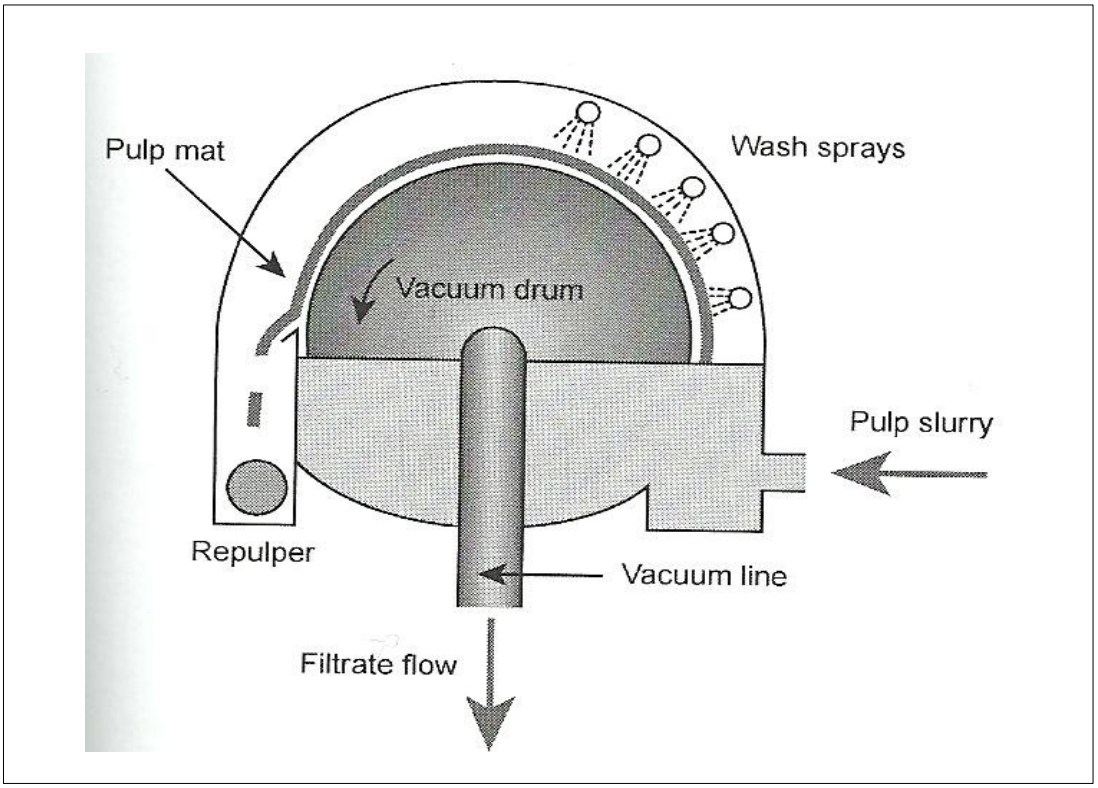
7.1
- Brightness
- Strength
- Yellowing
- Free from contaminants (4)

7.2
- Increase the mixer speed.
- Improve washing method between tower operations.
- Decrease load on the washer to allow for proper washing.
- Improve the mixing method to avoid channelling.
- Monitor concentration of chemical and time the pulp is exposed to chemicals. (5)

7.3 pH meter (1)

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7.4



(7)
[17]

QUESTION 8

8.1 8.1.1
- The paper roll unwinds as it is pulled through the rolls.
- The fly rolls tighten the paper before it enters the nip between the embossing rolls.
- The squeeze rolls pull the paper out of the embossing nip and make sure the pattern stays in place.
- The paper is rolled up. (4)

8.1.2
- The paper is rolled double and creases are caused.
- The design is missing on the product in certain areas.
- The design does not stay on the paper. (Any 2 × 1) (2)

8.2
- So that the paper looks and feels better✓
- To make printing easier✓
- To make processes like brushing and supercalendering easier
- To produce a high quality surface (Any 2 × 1) (2)

8.3 A de-aerator removes the trapped air by using hydrocyclones.✓ Hydrocyclones are types of separators that spin the fluid along its inner walls.✓ This motion forces air to move towards the centre.✓ Air is removed through the top of the hydrocyclones✓ while the fluid is removed from the bottom.✓ (5)

8.4
- Natural binders need to be cooked and can be attacked by bacteria. They need preservatives.
- Synthetic binders can be used directly and do not need to be cooked. No preservatives are needed. (2 × 2) (4)

[17]

TOTAL: 100

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